Orthogonal Projection.

Def. Let V be an inner product space, and $W<V$ be a subspace. Given a vector $\beta \in V$, if exists, a vector $\alpha \in W$ is called best approximation to β by vectors in W if:

$$\text{for all } \gamma \in W, \quad \|\beta-\alpha\| \le \|\beta-\gamma\| \quad \cdots (*)$$

Denote such a vector α as $\text{Proj}_W(\beta)$.

Thm. Let V be an inner product space and $W<V$ be a subspace. Given $\beta \in V$, and $\alpha \in W$,

$$\alpha = \text{Proj}_W(\beta) \text{ if and only if for all } \mu \in W, \ \langle \beta-\alpha, \mu \rangle = 0.$$

Proof. Notice that: for fixed $\alpha \in W$ and $\beta \in V$, given $\gamma \in W$,

$$\|\beta-\gamma\|^2 = \|\beta-\alpha+\alpha-\gamma\|^2 = \|\beta-\alpha\|^2 + \|\alpha-\gamma\|^2 + 2\cdot \text{Re}\langle \beta-\alpha, \alpha-\gamma \rangle$$

Therefore, the condition $(*)$ holds if and only if

$$\text{for all } \gamma \in W, \quad \|\alpha-\gamma\|^2 + 2\cdot \text{Re}\langle \beta-\alpha, \alpha-\gamma \rangle \ge 0. \quad \cdots (**)$$

Meanwhile, given $\mu \in W \setminus \{\alpha\}$, take

$$\gamma = \alpha + \frac{\langle \beta-\alpha, \alpha-\mu \rangle}{\|\alpha-\mu\|^2} \cdot (\alpha-\mu)$$

Then, substituting this to $(**)$,

$$|\langle \beta-\alpha, \alpha-\mu \rangle|^2 - 2\cdot |\langle \beta-\alpha, \alpha-\mu \rangle|^2 = -|\langle \beta-\alpha, \alpha-\mu \rangle|^2$$

Therefore, it satisfies ≥ 0 if and only if $\langle \beta-\alpha, \alpha-\mu \rangle = 0$, and since $\mu \in W \setminus \{\alpha\}$ was arbitrary, so the claim is proved.

Conversely, if $\beta-\alpha$ is orthogonal to every vector in W, then

$$\text{Re}\langle \beta-\alpha, \alpha-\gamma \rangle = \text{Re}\, 0 = 0,$$

Therefore, given $\gamma \in W$,

$$\|\beta-\gamma\|^2 = \|\beta-\alpha\|^2 + \|\alpha-\gamma\|^2 \ge \|\beta-\alpha\|^2,$$

So proved. ☑

Lemma. Let $\beta \in V$ be a vector and $W < V$ be a subspace.
If $\text{proj}_W(\beta) \in W$ exists, then it is unique.
Proof. Suppose that $\alpha_1, \alpha_2 \in W$ satisfy: for all $\gamma \in W$,

$$\langle \beta - \alpha_1, \gamma \rangle = \langle \beta - \alpha_2, \gamma \rangle = 0.$$

Then, by the linearity of inner product and take $\gamma = \alpha_1 - \alpha_2$,

$$\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle = 0, \quad \text{so } \alpha_1 = \alpha_2. \qquad \square$$

Def. Suppose that for all $\beta \in V$, there exists $\text{proj}_W(\beta)$. The map

$$P : V \to V : \beta \mapsto \text{proj}_W(\beta)$$

is called an orthogonal projection on W.

Thm. Every finite-dimensional Inner Product space has orthonormal basis.

Proof. Let V be an finite-dimensional inner Product space.

Let $\{\beta_1, \cdots, \beta_n\}$ be a basis for V.

Take $\alpha_1 = \beta_1$. And, for $2 \le m \le n$,

$$\alpha_m = \beta_m - \sum_{k=1}^{m-1} \frac{\langle \beta_m, \alpha_k \rangle}{\|\alpha_k\|^2} \cdot \alpha_k.$$

Then, for $n=2$, $\{\alpha_1, \alpha_2\}$ are orthogonal, because

$$\langle \alpha_2, \alpha_1 \rangle = \langle \beta_2, \alpha_1 \rangle - \frac{\langle \beta_2, \alpha_1 \rangle}{\|\alpha_1\|^2} \cdot \langle \alpha_1, \alpha_1 \rangle = 0.$$

And, suppose that for $2 \le m < n$, $\{\alpha_1, \cdots, \alpha_m\}$ are orthogonal. Then, for all $2 \le j \le m$,

$$\langle \alpha_{m+1}, \alpha_j \rangle = \langle \beta_{m+1}, \alpha_j \rangle - \sum_{k=1}^{m} \frac{\langle \beta_{k+1}, \alpha_k \rangle}{\|\alpha_k\|^2} \cdot \langle \alpha_k, \alpha_j \rangle$$

$$= \langle \beta_{m+1}, \alpha_j \rangle - \frac{\langle \beta_{k+1}, \alpha_j \rangle}{\|\alpha_j\|^2} \cdot \langle \alpha_j, \alpha_j \rangle = 0$$

By induction, the claim is proved. □

Thm. Let V be an inner product space, and let $W < V$ be a subspace. Suppose that: given $\beta \in V$, the orthogonal projection $\text{Proj}_W(\beta)$ exists. Then, the map

$$P : V \to V : \beta \mapsto \text{Proj}_W(\beta)$$

is linear, idempotent, hermitian map.

Proof. Notice that: $P(\beta)$ is a unique vector satisfying

for all $\gamma \in W$, $\langle \beta - P(\beta), \gamma \rangle = 0$. In other words, $\beta - P(\beta) \in W^\perp$.

1) Linearity.

Let $c \in F$ and $\alpha, \beta \in V$ be given. Then, for any $\gamma \in W$,

$$\langle c\cdot\alpha + \beta - (c\cdot P(\alpha) + P(\beta)), \gamma \rangle = c\cdot\langle \alpha - P(\alpha), \gamma \rangle + \langle \beta - P(\beta), \gamma \rangle = 0.$$

Thus $c\cdot P(\alpha) + P(\beta) = P(c\cdot\alpha + \beta)$, linear.

2). Idempotent.

Given $\beta \in V$, $P^2(\beta) = P(\beta)$, since $P(\beta) \in W$. Thus, $V = \text{Im}\, P \oplus \ker P$.

3). Hermitian.

Given $\alpha, \beta \in V$, $\alpha = P(\alpha) + (\alpha - P(\alpha))$ and $\beta = P(\beta) + (\beta - P(\beta))$.

Since $P(\alpha), P(\beta) \in W$ and $\alpha - P(\alpha), \beta - P(\beta) \in W^\perp$,

$$\langle P(\alpha), \beta \rangle = \langle P(\alpha), P(\beta) + (\beta - P(\beta)) \rangle = \langle P(\alpha), P(\beta) \rangle \quad \text{and}$$

$$\langle \alpha, P(\beta) \rangle = \langle P(\alpha) + (\alpha - P(\alpha)), P(\beta) \rangle = \langle P(\alpha), P(\beta) \rangle$$

Thus $\langle P(\alpha), \beta \rangle = \langle \alpha, P(\beta) \rangle$, therefore $P = P^*$.

4). $\text{Im}\, P = W$ and $\ker P = W^\perp$.

$\text{Im}\, P = W$ is clear by definition, and given $\alpha \in \ker P$,

$\alpha - P(\alpha) = \alpha \in W^\perp$. So $\ker P \subseteq W^\perp$. Conversely, given $\alpha \in W^\perp$,

$\alpha - 0 = \alpha \in W^\perp$, so $P(\alpha) = 0$, $W^\perp \subseteq \ker P$. ∎

Thm. Let $W<V$ be a finite dimensional subspace of inner product space V. Then, there exists the orthogonal projection onto W along $W^{\perp}$.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for W, using Gram-Schmidt.

Define $P: V \to V: \beta \mapsto \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \cdot \alpha_i$

Then, for each $1 \le \delta \le n$,

$$\langle \beta - P(\beta), \alpha_\delta \rangle = \langle \beta, \alpha_\delta \rangle - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \cdot \langle \alpha_i, \alpha_\delta \rangle = 0.$$

Thus proved. ㅁ

Thm. Let W be a fin. dim subspace of inn. prod space V. Then, the unique orthogonal projection on W exists. Further, P is idempotent linear operator with $\operatorname{Im} P = W$, $\ker P = W^\perp$.

Proof. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthonormal basis for W. Then, a map

$$P: V \to V: \beta \mapsto \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \cdot \alpha_i$$

1). It maps β to orthogonal projection by W. $\frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \alpha_i$ $\frac{\langle \gamma, \alpha_i \rangle}{\|\alpha_i\|^2}$.

For each $1 \le j \le n$,

$$\left\langle \beta - \sum_{i=1}^{n} \langle \beta, \alpha_i \rangle \alpha_i, \alpha_j \right\rangle = \langle \beta, \alpha_j \rangle - \langle \beta, \alpha_j \rangle = 0.$$

So, $\beta - P(\beta) \in W^\perp$ for all $\beta \in V$. (And, such a vector is unique, thus well-defined.)

2). It is linear.

Let $\beta, \gamma \in V$ and $c \in F$ be given. Then,

$$P(c\beta + \gamma) = \sum \langle c\beta + \gamma, \alpha_i \rangle \cdot \alpha_i = c \cdot \sum \langle \beta, \alpha_i \rangle \cdot \alpha_i + \sum \langle \gamma, \alpha_i \rangle \cdot \alpha_i = cP(\beta) + P(\gamma).$$

3). It is idempotent.

Given $\beta \in V$, $P(\beta) \in W$. So, $P^2(\beta) = P(\beta)$. ← projection

4). $\operatorname{Im} P = W$, $\ker P = W^\perp$.

$\operatorname{Im} P \subseteq W$ is clear, and for any $\gamma \in W$, $P(\gamma) = \gamma \in \operatorname{Im} P$. And,

$$\beta \in \ker P \iff P(\beta) = 0 \iff \langle \beta, \alpha_i \rangle = 0 \text{ for all } 1 \le i \le n \iff \beta \in W^\perp.$$

5). It is Hermitian.

Since $\operatorname{Im} P \oplus \ker P = V$, let $x_1, y_1 \in \operatorname{Im} P$ and $x_2, y_2 \in \ker P$ be given. Then

$$\langle P(x_1 + x_2), y_1 + y_2 \rangle = \langle x_1, y_1 + y_2 \rangle = \langle x_1, y_1 \rangle = \langle x_1 + x_2, y_1 \rangle = \langle x_1 + x_2, P(y_1 + y_2) \rangle.$$

□

Corollary. Let W be a fin. dim subspace of inn prod space V, and let P be an orthogonal projection on W. Then, $I - P$ is an orthogonal projection on $W^\perp$.

Proof. Given $\beta \in V$, $\langle \beta - P(\beta), \mu \rangle = 0$ for all $\mu \in W$, or $\beta - P(\beta) \in W^\perp$. So $\operatorname{Im}(I - P) = W^\perp$.

And, given $\gamma \in W^\perp$, $\beta - \gamma = P(\beta) + (\beta - P(\beta) - \gamma)$, therefore

($P(\beta) \in W$, $(\beta - P(\beta) - \gamma) \in W^\perp$)

$$\|\beta - \gamma\|^2 = \|P(\beta)\|^2 + \|\beta - P(\beta) - \gamma\|^2 \ge \|P(\beta)\|^2 = \|\beta - (\beta - P(\beta))\|^2 \text{ for all } W^\perp,$$

So $\beta - P(\beta)$ is the best approximation to β by vectors in $W^\perp$.

Note: let $E: V \to V$ be a projection. In general, E can not be determined by it's image.

For example, let $V = \mathbb{R}^3$, $W_1 = \text{Span}\{(1,0,0)\}$, $W_2 = \text{Span}\{(0,1,0),(0,0,1)\}$, $W_3 = \text{Span}\{(0,1,1),(1,1,1)\}$.

Then, $V = W_1 \oplus W_2 = W_1 \oplus W_3$, but $W_2 \neq W_3$. Moreover,

$E: V \to V: (x_1, x_2, x_3) \mapsto (x_1, 0, 0)$, $E': V \to V: (x_1, x_2, x_3) \mapsto (x_1 - x_3, 0, 0)$

Both E and E' are projections with $\text{Im}\, E = \text{Im}\, E' = \text{Span}\{(1,0,0)\}$ but $E \neq E'$,

$\ker E = W_2$ and $\ker E' = W_3$.

Corollary. Let $\{\alpha_1, \cdots, \alpha_n\}$ be an orthogonal set of non-zero vectors in inn.prod space V. Then,

for all $\beta \in V$, $\sum_{i=1}^{n} \frac{|\langle \beta, \alpha_i \rangle|^2}{\|\alpha_i\|^2} \leq \|\beta\|^2$, the equality holds iff $\beta = \sum \frac{\langle \beta, \alpha_i \rangle}{\|\alpha_i\|^2} \cdot \alpha_i$.

Prof. Take $\gamma = \sum [\langle \beta, \alpha_i \rangle / \|\alpha_i\|] \cdot \alpha_i$. Then, $\langle \beta - \gamma, \gamma \rangle = 0$. So

$$\|\beta\|^2 = \|\gamma\|^2 + \|\beta - \gamma\|^2 \geq \|\gamma\|^2.$$

Thm. Let W be a finite-dimensional subspace of an inn. Prod space V.
If $P: V \to V$ is an orthogonal projection on W, that is,

for any $\beta \in V$, $\|\beta - P(\beta)\| \leq \|\beta - \gamma\|$ for all $\gamma \in W$ ... (*)

then, P is idempotent hermitian linear map s.t $\operatorname{Im} P = W$, $\ker P = (\operatorname{Im} P)^{\perp}$, $V = \operatorname{Im} P \oplus \ker P$

Proof. Since $P(\beta) \in W$, $P(P(\beta)) = P(\beta)$, for all $\beta \in V$, so $P^2 = P$, idempotent.

To show the linearity, let $\alpha, \beta \in V$ and $c \in F$ be given. Since $\alpha - P(\alpha)$, $\beta - P(\beta) \in W^{\perp}$, so

$$c \cdot (\alpha - P(\alpha)) + (\beta - P(\beta)) = (c\alpha + \beta) - (cP(\alpha) + P(\beta)) \in W^{\perp}.$$

So, $cP(\alpha) + P(\beta) = P(c\alpha + \beta)$, linearity is obtained

To show the direct sum, given $\beta \in V$, $\beta = P(\beta) + (\beta - P(\beta))$, and since $\beta - P(\beta) \in W^{\perp}$, $V = \operatorname{Im} P + W^{\perp}$ is obtained. And,

$\alpha \in W^{\perp} \Leftrightarrow P(\alpha) = 0$, because

$\alpha \in W^{\perp} \Leftrightarrow \forall \mu \in W$, $\langle \alpha, \mu \rangle = 0 \Leftrightarrow P(\alpha) = 0$. So, $V = \operatorname{Im} P + \ker P$. And, $\operatorname{Im} P \cap \ker P = \{0\}$,

$\langle \alpha - 0, \mu \rangle = 0$.

because $P(\alpha) = 0$ and $\alpha = P(\alpha)$ implies $\alpha = 0$.